Shih 2018 ovarian cancer — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=20) collapses the strong patient batch effect: same-sample neighbour fraction 0.73→0.20 (perfect-mixing floor 0.16), iLISI median 2.98→5.82 of 7 samples. One connected component (0 islands). Patient islands (pre-Harmony) fuse into one continuous mixed cloud.

Gate 2 — module geography → GO (non-branching)

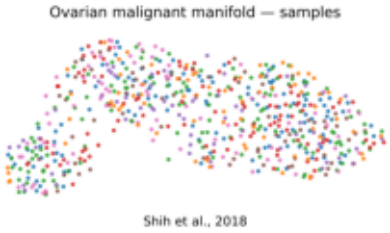
Single (Gast/pSom)→Crest developmental gradient. Gast & pSom co-vary (smoothed r=+0.23; joint enrichment 0.95 ~ origin overlap), both oppose Crest (Gast↔Crest r=−0.80, pSom↔Crest r=−0.74). Along pseudotime Gast/pSom fall (ρ −0.86/−0.62), Crest rises to a terminal tip (decile pt=0.99); joint pSom&Crest enrichment 0.26 (<1 = inverse gradient, not a hub).

Gate 3 — cross-repression → NA

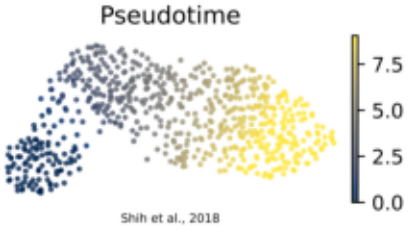
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks; PPT tree 0 branch nodes). No bifurcation, so the branching-only hub cross-repression test is not run.

STEP FIGURES

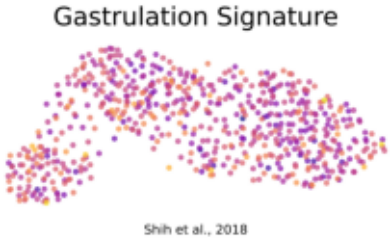
Manifold — samples



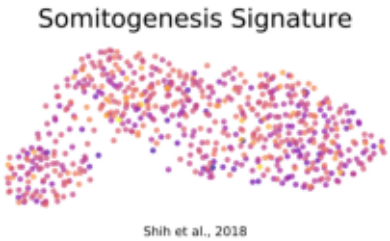
Pseudotime



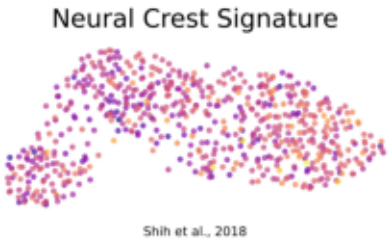
Gastrulation



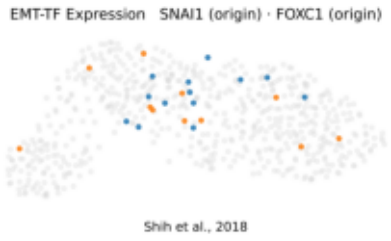
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
HGSOC/LGSOC ovarian tumours (Shih et al., 2018), SmartSeq2 — 633 malignant cells over 7 samples; sparse (~1250 genes/cell), a single low-cycling body with a cycling spur (~43% cycling). Non-branching Linear-pSom/Crest.

Verdict.
Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous gastrulation/somitogenesis → neural-crest axis, Crest terminal. G1 GO / G2 GO / G3 NA. EMT-TFs SNAI1/FOXC1/PRRX2 all origin/root-enriched (no terminal EMT-TF).

Parameter choices

Sample threshold	floor ≥100 rejected (3/16); cliff 60→37 → geo-mean 47 → 7/16, 633 cells	**Feature selection**	scFates HVG underpowered on 633 sparse cells (33) → Seurat HVG 2000
Integration	Harmony on sample θ=20 (mixing collapses to floor; strong ovarian batch)	**Embedding**	n_pcs=20, n_neighbors=30, spread=1.3 (SS2 precedent); 0 islands removed
Geometry	non-branching curve N=20, λ=0.3, μ=100; 2 tips / 0 forks (2.84)	**Root**	tip 2 = Gast/pSom-high, Crest-low origin; axis cell-cycle-free (ρ=−0.03)
Module means (raw)	Gast −0.013 · pSom +0.065 · Crest −0.034 (Linear-pSom/Crest)	**EMT-TFs**	SNAI1 (8.4%), FOXC1 (6.8%), PRRX2 (3.5%) — all origin/root-enriched; overlay SNAI1 & FOXC1